

Validation-Lot Purchase Order and Release Specification

Purchase Order BE-PO-240819-2147

Issued by Blue Ember Biologics, LLC on August 19, 2024 to Granite Heron Logistics, Inc. The attached litigation copy was filed with Blue Ember's complaint on December 2, 2024 in Civil Action No. SYN-WDNC-24-CV-0520.

Blue Ember orders one dedicated deep-frozen movement of BE-214 validation lot BE214-VL-2409A from Solstice Gate Analytics, 61 Pelham Lab Way, Greenville, South Carolina, to Blue Ember, 1847 Cinder Ridge Parkway, Asheville, North Carolina. Requested pickup is September 12, 2024, with same-day delivery and a minus-seventy-degree-Celsius trailer setpoint.

The shipment is governed by Master Logistics Services Agreement BE-GHL-MLS-2023-1101, Quality Agreement QA-BE-GHL-017, Statement of Work SOW-BE214-VAL-04, and Release Specification BE214-RS-03, Revision 3. Acceptance of this order does not permit Granite Heron to substitute standard tariff terms for the signed temperature or custody controls.

Blue Ember's procurement authorization confirms budget and ordering authority. Quality acceptance remains separate and depends on compliant handling, intact seals, complete records, and review of the designated logger.

The order does not purchase manufacturing judgment from the carrier. It purchases documented transport performance for one identified lot, while leaving Blue Ember responsible for scientific disposition and Granite Heron responsible for the equipment and custody steps it accepts.

Ordered line items and commercial terms

Line one is dedicated controlled transportation for the Greenville-to-Asheville lane, including qualified trailer service, origin and destination appointments, continuous refrigeration, ordinary Fletcher staging, and the agreed record packet. Line two is the enhanced declared-value option for the validation payload. Line three covers the CryoTrace-compatible monitoring and data-export service.

Granite Heron will state the final line-haul price in Rate Confirmation RC-240912-087.

Accessorial charges require a contemporaneous supporting event and may not be created from an unrecorded delay. Invoice terms are thirty days after complete delivery of the service file.

Blue Ember may withhold a genuinely disputed item while paying unrelated accepted charges.

The declared payload value is 1.35 million dollars for cargo-limit purposes. That figure reflects cost embodied in the validation lot; it is not an agreed liquidated-damages amount and does not include replacement work or schedule consequences. The selected protection displaces a lower default cargo ceiling only for proven direct payload loss.

Granite Heron shall reference the purchase order, lot, shipment number, and service date on its invoice. Payment or processing of a transportation charge does not waive a concealed quality deviation. A claim does not excuse Blue Ember from paying an undisputed service component.

Shipment instructions and delivery conditions

Granite Heron shall use dedicated tractor GH-4821 and trailer TR-7714 unless an emergency replacement of equal capability is documented. The trailer target is minus seventy degrees Celsius, and personnel must preserve the authorized range throughout custody. Temporary staging at the Fletcher facility is permitted only under active refrigeration.

The payload logger is CryoTrace CTX-9 serial CTX9-44721. Solstice Gate Analytics will activate and place it, while the driver will verify the serial at handoff. All September 12 entries use Eastern Daylight Time. The carrier must preserve the original export and provide a readable report after delivery.

Urgent exceptions are reported by telephone through the controlled contact sheet, beginning with Dr. Mira Solis. If the primary contact does not answer within ten minutes, personnel use the secondary route and continue reasonable protective action. Restoration of temperature does not cancel the notification requirement.

Delivery requires an intact custody handoff at Blue Ember's Asheville receiving area. The recipient may sign for physical possession while marking the lot quarantined. Only the Blue Ember quality unit may approve use after an excursion. No driver, dispatcher, or procurement employee may waive the release specification orally.

Release Specification BE214-RS-03, Revision 3

This release specification became effective August 1, 2024 and governs validation lot BE214-VL-2409A. Its purpose is to ensure that material used for BE-214 process validation has both acceptable test attributes and documented environmental custody. Revision 3 replaces Revision 2 for shipments occurring on or after the effective date.

The lot identifier must match the certificate of conformance, custody form, laboratory samples, and campaign record. Seals must be intact absent an authorized inspection. The required transport range is minus eighty through minus sixty degrees Celsius from carrier acceptance to destination receipt. The nominal transportation setpoint is minus seventy.

No time-based temperature tolerance is approved for this validation lot. A value outside the stated range triggers quarantine and deviation review. A boundary reading within the range is acceptable, while a reading warmer than minus sixty or colder than minus eighty is out of specification. Rounding may not convert an unrounded failing value into a passing value.

This document does not state that every excursion chemically destroys every vial. It states that compliant custody is a release attribute for the intended validation use and requires a documented scientific decision after any failure.

Release attributes and evidence

Attribute one is identity: records and labels must consistently show BE214-VL-2409A. Attribute two is container integrity: seals, closures, and vials must show no unexplained breach. Attribute three is temperature compliance based on the designated logger. Attribute four is documentation completeness, including custody times, equipment identity, stop chronology, and exception records.

Attribute five is confirmatory analytical performance under the approved BE-214 methods. Results are evaluated against the method-specific limits in the controlled laboratory annex. The annex contains proprietary technical ranges not reproduced in the transport copy. Analytical conformity does not automatically cure failure of the validation-custody attribute.

The reviewer shall use the raw CTX-9 export, not merely a carrier summary. Trailer data may corroborate timing and mechanism. If clocks differ, the assessment must document the offset rather than move source readings without explanation. An unexplained data gap is itself reviewed for impact.

Release requires an authorized quality signature. Receiving personnel may place material in frozen quarantine but cannot declare it conforming. Procurement approval, carrier delivery, or payment of freight is not quality release. Each rejected attribute must identify the evidence and the relationship to the lot's intended use.

Sampling, testing, and stability assessment

After a reported excursion, Blue Ember may request representative sampling and confirmatory testing. Samples must remain traceable to the lot and be handled under a protocol that avoids adding a second uncontrolled exposure. The laboratory will record method, analyst, instrument, date, result, and any deviation.

Stability assessment considers magnitude, duration, thermal recovery, packaging configuration, available development data, and the planned validation role. A passing activity result may show that tested samples retain a measured function. It may not establish uniform condition across all vials or reconstruct documentation that the validation protocol required.

Conversely, the quality reviewer shall not claim total biochemical failure when testing does not support that conclusion. The disposition rationale should identify whether rejection rests on measured inactivity, uncertainty, failure of a validation prerequisite, or a combination. Scientific precision is required even when the ultimate decision is rejection.

Granite Heron may receive a sufficient summary for claim evaluation, subject to protection of formulation and method details. Carrier comments should be considered but do not transfer release authority. Retained samples and source data will be preserved under the applicable development schedule and any later litigation hold.

Deviation disposition and replacement planning

An out-of-range reading places the lot in quarantine. Blue Ember will open a deviation, obtain carrier records, review analytical information, and select release, conditional use, additional testing, nonvalidation use, rework, return, or destruction. The decision must be dated, signed, and supported.

If the lot cannot serve its validation purpose, the program team will evaluate replacement capacity. The assessment should identify available facilities, earliest credible dates, transfer and qualification tasks, change-control requirements, direct cost, technical risk, and schedule effect. Quality safeguards remain mandatory, but review should proceed promptly enough to avoid preventable delay.

An alternative site need not be previously approved to be considered. Nor does open capacity establish that the site is ready. A reasonable decision weighs the work remaining and information obtainable before the offer expires. Assumptions and unresolved questions should be recorded contemporaneously.

Nothing in this specification decides the legal mitigation burden or fixes a damages deduction. It creates the quality record from which business and legal decisions may later be evaluated. Any rejected slot, retained reservation, or replacement campaign should preserve supporting communications, budgets, and timing evidence.

Authorization and revision history

Revision 3 clarifies that the transport band has no authorized duration tolerance for validation lot BE214-VL-2409A, distinguishes analytical activity from the validation-custody attribute, and adds explicit documentation for alternative-site review. No change was made to the minus-seventy setpoint or the minus-eighty to minus-sixty acceptance band.

Dr. Mira Solis approved the revision for Blue Ember on August 1, 2024. The controlled copy was supplied to Granite Heron before execution of the August 19 statement of work. Adrian Locke acknowledged receipt for the carrier; acknowledgment confirms access to the instruction and does not give Granite Heron authority to alter it.

Purchase order BE-PO-240819-2147 was issued on August 19, 2024. The release specification's earlier effective date remains August 1. Both were included in the complaint materials filed December 2, 2024, which did not change either underlying date.

Specification approval

Dr. Mira Solis, Blue Ember Biologics, LLC

Effective August 1, 2024

Carrier receipt acknowledgment

Adrian Locke, Granite Heron Logistics, Inc.

Received August 19, 2024

All names, signatures, entities, numbers, and scientific descriptions in this exercise instrument are fictional.